Predictive Modeling Plan: Delinquency Risk Forecasting for Geldium

> **Project:** AI-Driven Delinquency Risk Assessment for Geldium's Credit Card Collections
> **Document 2 of 4:** Predictive Modeling Plan

Objective

Build a model that predicts the **probability that a customer will become delinquent** on their credit card payments, so the Collections team can intervene early. The model must be **accurate**, but also **interpretable, fair, and acceptable to regulators** — the financial domain demands that every risk decision can be explained.

Step 1: Predictive Model Logic

1.1 Chosen Approach: A Dual-Model Strategy

	Baseline Model	Advanced Model
Algorithm	Logistic Regression	Gradient Boosted Trees (e.g., XGBoost)
Main strength	Interpretability, ease of deployment, transparency	Higher predictive performance; handles non-linear feature interactions
Role	Trusted, explainable baseline	Performance upgrade, explained with SHAP

Because finance places heavy emphasis on interpretability and fairness, logistic regression serves as the trusted baseline. To increase predictive power, a second version is built using gradient boosting.

1.2 Key Input Features

Selected based on the EDA findings — the top 5 features for modelling:

#	Feature	Why It Matters
1	**Credit Utilisation Ratio**	High ratios (>90%) strongly correlate with delinquency
2	**Payment History (3-month window)**	Recency and frequency of missed payments are strong predictors
3	**Debt-to-Income Ratio**	Indicates financial stress and repayment capacity
4	**Auto-Pay Enrollment Status**	Customers not on auto-pay miss payments more often
5	**Employment Change Indicator**	Sudden changes in employment can disrupt income stability

1.3 Model Workflow

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Data Ingestion → Feature Engineering → Model Training → Prediction Output → Thresholding & Classification
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1. **Data Ingestion** — The cleaned, pre-processed dataset is fed into the pipeline.
2. **Feature Engineering** — Derived ratios, binary indicators, and normalised continuous variables are created.
3. **Model Training** — Train the baseline logistic regression first, then a hyperparameter-tuned gradient boosting model.
4. **Prediction Output** — The model outputs a **probability score between 0 and 1** representing delinquency risk.
5. **Thresholding & Classification** — Customers above a risk threshold (e.g., **0.65**) are **flagged for intervention**.

Step 2: Model Justification

Why Logistic Regression as the Primary Model

- **Interpretability:** Coefficients directly show how each feature raises or lowers risk.
- **Regulatory acceptability:** Financial institutions often need transparent models that explain *why* a decision was made — especially when it affects credit and collections.
- **Easy to explain to business stakeholders:** Probability outputs translate naturally into explainable risk scores.

Why Add Gradient Boosting (XGBoost)

- Captures **complex, non-linear relationships** between features.
- Typically **outperforms linear models** on structured financial data.
- It is less interpretable by default, but **SHAP (SHapley Additive exPlanations)** can explain individual predictions, keeping a level of transparency.

Strategic Fit

The dual-model approach supports Geldium's goals of **improving prediction accuracy** while maintaining **fairness** and **operational feasibility**.

Step 3: Evaluation Strategy

3.1 Metrics for Accuracy and Reliability

Metric	What It Measures	Why It's Used
AUC-ROC	Ability to distinguish delinquent from non-delinquent customers	Threshold-independent measure of ranking quality
F1 Score	Balance between precision and recall	Important because the data is imbalanced (few delinquents)
Accuracy	Overall correctness of classification	Used with caution, since class imbalance can inflate it
Confusion Matrix	True/false positives and negatives	Shows how many risky customers are missed (false negatives)

3.2 Fairness & Bias Detection

- **Demographic Parity & Equal Opportunity checks** — ensure the model does not discriminate across sensitive groups such as age, gender, or income segment.
- **Disparate Impact Ratio** — tests whether any demographic group is disproportionately flagged as high risk.
- **SHAP Value Analysis** — identifies which features drive risk predictions, ensuring there is no hidden bias in feature influence.

3.3 Model Monitoring

- Track performance over time using a validation set and **monitor for model drift**.
- **Re-evaluate the model monthly** using updated customer data.
- Set up a **feedback loop with the Collections team** for misclassified cases.

Conclusion

The modelling framework is designed to meet Geldium's strategic objectives:

- **Better delinquency detection**
- **Guided intervention strategies**
- **Ethical AI practices**

By combining **logistic regression's transparency** with **gradient boosting's accuracy**, the plan offers a balanced, effective, and responsible approach to credit risk modelling. Evaluation and fairness checks are built into the framework to support ongoing model governance and business confidence.

At a glance

- Models: Logistic Regression (baseline) + XGBoost (advanced, explained via SHAP)
- Output: risk probability 0–1; flag threshold ≈ 0.65
- Top features: utilisation, 3-month payment history, DTI, auto-pay, employment change
- Metrics: AUC-ROC, F1, accuracy, confusion matrix
- Fairness: demographic parity, equal opportunity, disparate impact ratio, SHAP
- Monitoring: monthly re-evaluation, drift tracking, Collections feedback loop